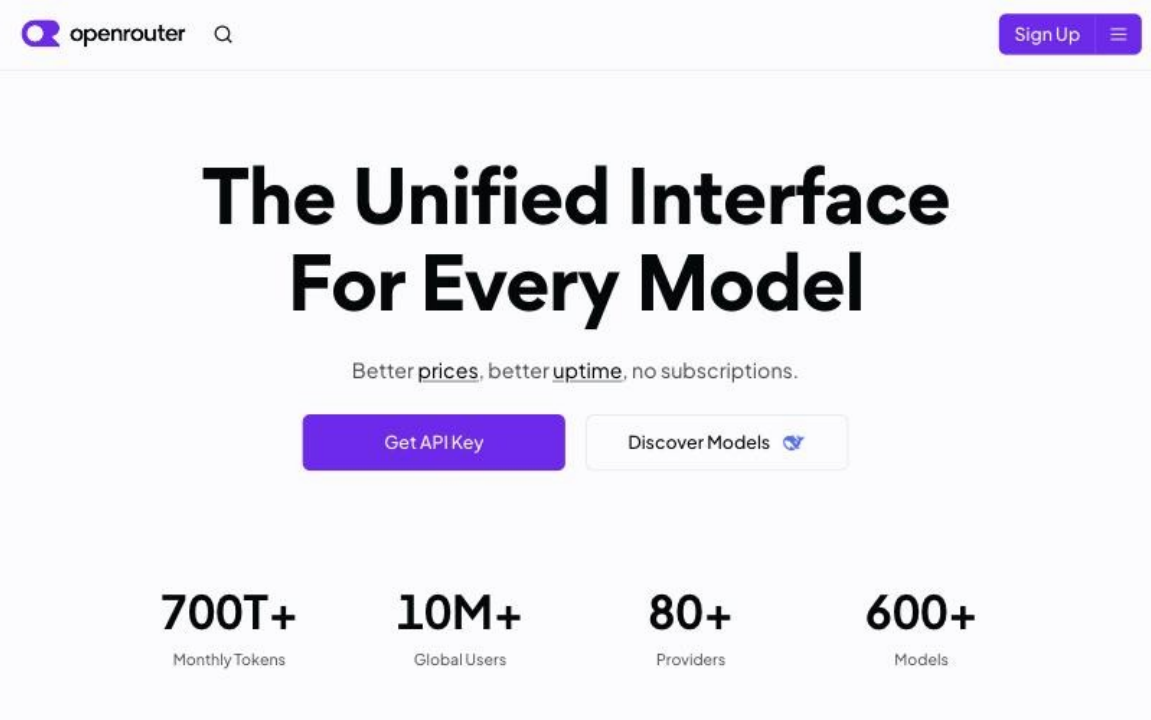
How to Create an OpenRouter Account and Get an API Key - v2

Version 2. Screenshots were captured on 2026-11-10 from the public openrouter.ai site. Nothing was entered in any form and no account was created for this guide, so the steps after sign-up (credits and key creation) are written from OpenRouter's documentation and are not shown in screenshots. The test program in section 6 was not run with a real key. Sites change; check the on-screen wording. This guide has no dependency on any workshop files.

1. What is OpenRouter?

OpenRouter is a service with one account and one API key for many AI models from different providers (it advertises hundreds of models and no subscriptions). Programs send requests to OpenRouter's address and choose a model by name. You pay for what you use with prepaid credits, and some models are free (see section 5).



openrouter.ai home page, signed out. The Sign Up button is at the top right.

2. Before you start

- This is optional unless your instructor asks for it. If your workshop provides a key, use that one instead and do not buy credits.

- An API key works like a password for your account: whoever has it can spend your credits. Never share it, post it, or put it in a notebook, slide or screenshot.
- Use your own account. Do not share one key between people.
- Free models need no payment, but they have low daily and per-minute limits.

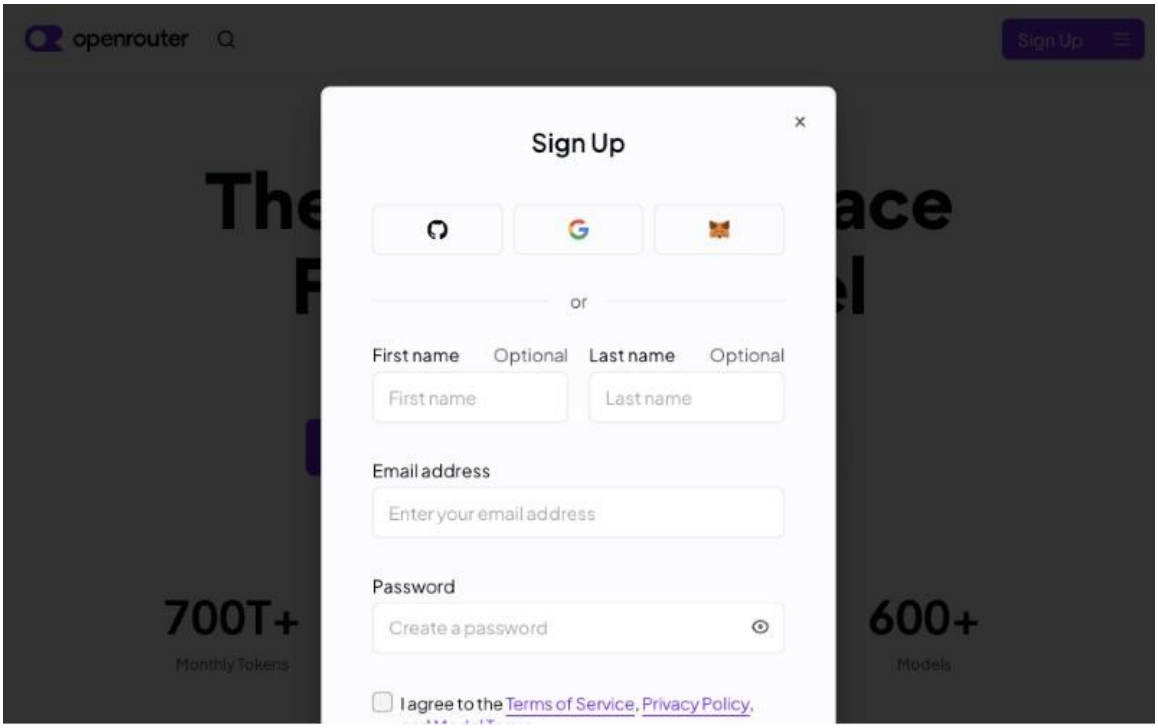
3. Create an account

1. Go to openrouter.ai and click Sign Up (top right).

2. Choose how to sign up: continue with GitHub, with Google, or with a crypto wallet (MetaMask), or fill in your email address and a password. First and last name are optional.

3. Tick the box to agree to the Terms of Service and Privacy Policy, then continue. Read them first; the sign-up form links to them.

4. If you used email, you may be asked to verify your email address. Open the message and follow the link.



The Sign Up dialog (left empty). Use one of the three buttons at the top or the email form below.

4. Create an API key

1. Sign in and open openrouter.ai/settings/keys (or click your profile menu and choose Keys). The home page Get API Key button also leads there.

2. Click Create API Key (or Create Key).

3. Give the key a clear name, such as "workshop-laptop". Set a credit limit for the key (for example a few dollars). A limit protects you if the key leaks. If you only plan to use free models you can try a very low limit; if free requests are then refused, raise it slightly or ask your instructor, because this behaviour was not tested for this guide.

4. Create the key and copy it right away. Keys are normally shown in full only when they are created, so paste it into a password manager or a secure place immediately.

Keep the key out of your code. Keys created during OpenRouter onboarding may start with a spending limit and expiry date; check the key settings page.

Where to store the key

On your own computer, store it as an environment variable named OPENROUTER_API_KEY and have your program read it (see the first test program in section 6).

```
# macOS / Linux (current Terminal session only)
export OPENROUTER_API_KEY="paste-your-key-here"

# Windows PowerShell (current session only)
$env:OPENROUTER_API_KEY = "paste-your-key-here"
```

In Google Colab, saving a key in the Secrets pane does NOT create an environment variable. Do this instead:

1. Click the key icon in the left bar of the notebook and choose Add new secret.

2. Name it OPENROUTER_API_KEY, paste the key as the value, and switch on Notebook access for the notebook you are using.

3. Read it in code with userdata.get (second test program in section 6). Never print the value.

Notebook access lets the code that runs in that notebook read the key, so only grant it to notebooks you wrote or have reviewed, and never share a notebook that you have granted access to without checking it first.

5. Credits, free models and limits

- Credits are US-dollar deposits used for requests. Add them at openrouter.ai/settings/credits. Cards, AliPay and USDC are accepted. Purchases may take a while to appear, and unused credits may expire after one year. Fees and refund rules are on the site.
- Free models have names ending in :free and have their own limits. According to OpenRouter's documentation, accounts without purchased credits get about 50 free-model requests per day; with at least 10 credits purchased the limit is about 1,000 per day. There is also a per-minute limit (documented as about 20 requests per minute at the time of writing). The model name openrouter/free picks an available free model for you.
- Limits and provider availability change; the OpenRouter limits page is the authority.
- You do not need to add credits to try the free models, and you should not buy credits just to complete a readiness check.

6. Test your key (optional)

Install the OpenAI Python package (python -m pip install openai). OpenRouter accepts the same request format. A printed reply means your key works. Each statement below is short so that it copies cleanly from this document; the same code is also provided as a separate file, test_openrouter.py. Check that every line is copied exactly, including the indentation of the continuation lines.

A. On your computer (key in an environment variable):

```
import os
from openai import OpenAI

URL = "https://openrouter.ai/api/v1"
KEY = os.environ["OPENROUTER_API_KEY"]
client = OpenAI(base_url=URL, api_key=KEY)

msgs = [{"role": "user", "content": "Say hello."}]
reply = client.chat.completions.create(
    model="openrouter/free", messages=msgs)
print(reply.choices[0].message.content)
```

B. In Google Colab (key in Secrets, with notebook access switched on):

```
from google.colab import userdata
from openai import OpenAI

URL = "https://openrouter.ai/api/v1"
KEY = userdata.get("OPENROUTER_API_KEY")
client = OpenAI(base_url=URL, api_key=KEY)

msgs = [{"role": "user", "content": "Say hello."}]
reply = client.chat.completions.create(
    model="openrouter/free", messages=msgs)
print(reply.choices[0].message.content)
```

If the openai package is missing in Colab, run !pip install openai in a cell first. These snippets were written from the documentation and were not run with a real key for this guide.

7. Privacy and safety

- OpenRouter says it records basic request details (such as time, model and token counts) and does not store your prompts and answers by default; logging them is an opt-in setting. This is not a guarantee that nothing is ever retained: OpenRouter's documentation describes exceptions, for example legal or abuse-review preservation, and some endpoints have their own retention rules. Each model provider also has its own logging and training policies. Read the current data-collection and provider-logging pages before sending anything you care about.
- Send only data you are allowed to share. Do not send patient, identifiable, confidential or unpublished data to any AI service.
- Never commit keys to GitHub or paste them into chats, notebooks or screenshots.
- If a key is exposed or you are unsure, delete it on the keys page and create a new one right away. OpenRouter may email you if it detects a leaked key.

8. Troubleshooting

Read the full error message first; it usually says what is wrong.

- KeyError: OPENROUTER_API_KEY (before any request is sent): the program cannot find the key. On your computer, set the environment variable in the same terminal you run the program in. In Colab, use the userdata method in section 6B; Secrets do not become environment variables.
- 401 Unauthorized: the key is wrong, deleted, expired or not being passed; check how it is read, then create a new key if needed.
- 402 Payment Required: the key or account has no available credit, or the key reached its credit limit. Check the balance and the key limit on the website before changing anything; you do not have to pay to complete a readiness check.
- 429 Too Many Requests: you hit a daily, per-minute or provider-side limit. Wait and retry; if the response gives a Retry-After time, wait at least that long, and increase the wait between repeated retries. Free models share limits and can be busy.
- Cannot sign in or the page does not load: try another browser or disable extensions that block scripts.
- Wrong model name: copy the exact model name from openrouter.ai/models.